

P1 - Academic Document Field Classification and Extraction

Updated Build Plan for CONSTANT, CONTROLLED_VARIABLE, and OPEN_VARIABLE
Detection

Prepared for Academic Document Intelligence Project

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1. Problem Statement

P1 is an academic document field-classification and extraction system. Given a scanned or digital university document, the system identifies which visible content is fixed template text and which content represents document-specific fields.

The target MVP focuses on:

- One institution, for example JNTUH or a single selected university.
- Two document types:
 - Degree certificate.
 - Marksheet / transcript-style document.
- 50-100 real documents collected and annotated for the first sprint, expanding to 100-300 documents after the baseline is working.

The system must classify and extract fields such as:

- University name.
- Degree name.
- Branch name.
- Student name.
- Hall ticket / registration number.
- Marks / grades / CGPA.
- Subject codes.
- Semester / regulation.
- Issue / exam dates.
- Seals, signatures, logos, and other structural regions.

The output should be a versioned JSON object that can be reviewed, audited, and used by downstream verification systems.

2. Core Design Principle

Do not start with deep learning only. The first useful system should be a template-aware, OCR-based baseline.

The baseline should:

1. OCR all documents.
2. Normalize layouts into a shared coordinate system.
3. Cluster OCR boxes by position and text similarity.
4. Identify repeated text as template content.
5. Identify non-repeated values as candidate fields.
6. Use this baseline as both an MVP and a weak-label generator for later model training.

This gives a working system quickly and prevents the project from depending entirely on a large model that may not converge within the sprint.

3. Revised Label Taxonomy

The original binary taxonomy, CONSTANT vs VARIABLE, is not enough for real academic documents. Many important fields change across documents but come from a controlled set of possible values. Therefore, the final taxonomy should use three primary field classes.

3.1 Primary Classes

Class	Meaning	Examples	Validation Method
CONSTANT	Repeated template text that should remain almost identical across documents of the same template	University name, fixed legal declaration, fixed footer	Repetition rate and template match
CONTROLLED_VARIABLE	Changes across documents but belongs to a known finite list	Degree name, branch, semester, regulation, subject code	Lookup table / controlled vocabulary
OPEN_VARIABLE	Free-form or open-ended field	Student name, hall ticket number, CGPA, marks, DOB	Pattern validation, OCR confidence, human review

3.2 Supporting Region Labels

Additional labels are useful for region-level understanding:

- TABLE_HEADER.
- TABLE_CELL.
- LOGO.
- SEAL.
- SIGNATURE.
- PHOTO.
- BACKGROUND.
- NOISE / ARTIFACT.

3.3 Why CONTROLLED_VARIABLE Matters

A field such as **Bachelor of Technology** should not be treated the same way as a student name. It changes across documents, but it belongs to a known list of valid degree names.

Examples:

Field	Correct Class	Reason
University name	CONSTANT	Same on most documents
Bachelor of Technology	CONTROLLED_VARIABLE	Changes across programs but comes from a known list
Computer Science and Engineering	CONTROLLED_VARIABLE	Branch names are controlled values
A. Ravi Kumar	OPEN_VARIABLE	Free-form name
8.72	OPEN_VARIABLE	Numeric value with pattern validation

This reduces annotator disagreement and allows automatic correction or review using lookup lists.

4. MVP Scope

The MVP must stay narrow.

Dimension	MVP Scope
Institution	1 institution
Document types	Degree certificate + marksheet
Templates	2-5 known template versions
Documents	50-100 real docs initially, then 100-300
Languages	English first, but benchmark Indic-script OCR if present
Output	Structured JSON with schema_version
Review	Human review for low-confidence fields

5. Data Collection and Annotation Plan

The 6-week roadmap must explicitly include data collection. The project will stall if annotated documents and reference templates are assumed to exist.

5.1 Real Document Collection

Collect:

- 30-50 degree certificates.
- 20-50 marksheets.
- At least 2-3 template versions if available.
- Multiple scan conditions:
 - clean PDF,
 - phone photo,
 - low-light photo,
 - photocopy,
 - compressed WhatsApp-style image,
 - skewed scan.

Minimum Week-1 target:

- 50 real documents ingested.
- 20 documents manually annotated.
- 5-10 marksheets with tables annotated.

Expanded target:

- 100-300 documents after MVP baseline is functional.

5.2 Annotation Objects

Each annotation should include:

- Bounding box.
- OCR text.
- Label class.
- Field name.
- Field value.
- Source document ID.
- Template ID.
- Annotator ID.
- Review status.

Example annotation:

```
{
  "field_name": "student_name",
  "label_class": "OPEN_VARIABLE",
  "text": "A. Ravi Kumar",
```

```

"bbox": [241, 318, 468, 351],
"document_id": "DOC_00042",
"template_id": "JNTUH_CERT_V2",
"annotator_id": "ann_01",
"review_status": "verified"
}

```

5.3 Annotation Rules

Rule 1 - Do not over-label entire sentences

If the template sentence is:

This is to certify that A. Ravi Kumar has completed the course.

Only A. Ravi Kumar should be OPEN_VARIABLE. The rest is CONSTANT.

Rule 2 - Controlled vocabulary fields get CONTROLLED_VARIABLE

Examples:

- Bachelor of Technology.
- Master of Technology.
- Computer Science and Engineering.
- Semester I.
- Regulation R18.

Rule 3 - Tables must be annotated cell-wise

For marksheets, annotate:

- table title,
- column headers,
- row labels,
- subject codes,
- subject names,
- internal marks,
- external marks,
- total marks,
- grade,
- credits.

6. Shared Preprocessing and OCR Core

P1 should use a shared document intelligence core so that P2 can reuse it later.

6.1 Input Types

Supported inputs:

- PDF.
- JPEG.
- PNG.
- TIFF.
- Scanned image.
- Phone-camera image.

6.2 Preprocessing Steps

INPUT DOCUMENT

- > file validation
- > page/image extraction
- > orientation detection
- > deskewing
- > denoising
- > contrast normalization
- > resolution normalization
- > quality scoring
- > OCR
- > layout normalization
- > candidate field generation

6.3 Quality Score

Quality score should capture:

- blur,
- compression,
- skew,
- low contrast,
- glare,
- shadow,
- partial crop,
- low resolution,
- photocopy degradation.

Quality score should be logged as a calibration feature, not just a rejection condition.

7. OCR Strategy

7.1 OCR Engines to Benchmark

Benchmark these OCR options:

- PaddleOCR.
- TrOCR.
- Tesseract with Indic language packs.
- Any institution-specific OCR fine-tune if data becomes available.

PaddleOCR is a strong practical default because its ecosystem includes OCR and document-structure modules, including table-structure recognition through PP-Structure / table recognition components. The official PaddleOCR documentation describes table structure recognition as converting non-editable table images into editable formats such as HTML by identifying rows, columns, and cells.¹

7.2 OCR Benchmark Table

OCR Engine	English Text CER	Indic Text CER	Table Text Accuracy	Seal/Header Accuracy	Speed	Notes
PaddleOCR	TBD	TBD	TBD	TBD	TBD	Primary default
TrOCR	TBD	TBD	TBD	TBD	TBD	Useful on printed text

¹PaddleOCR table-structure-recognition documentation: https://paddlepaddle.github.io/PaddleOCR/main/en/version3.x/module_usage/table_struct_recognition.html

OCR Engine	English Text CER	Indic Text CER	Table Text Accuracy	Seal/Header Accuracy	Speed	Notes
Tesseract Indic	TBD	TBD	TBD	TBD	TBD	Useful baseline for Indic scripts

7.3 Indic-Script Handling

Indian university documents may contain Telugu, Hindi, Sanskrit, or other Indic text in seals, headings, and signatures. The system should not assume English-only OCR is sufficient.

Benchmark separately:

- English printed text.
- Indic printed text.
- Seal text.
- Curved / circular text.
- Low-resolution header text.

8. Template-Differencing Baseline

The baseline is the first system to build.

8.1 Algorithm

```

FOR each document:
  OCR the page
  Normalize all OCR boxes to page size
  Store text, bbox, font/visual cues, OCR confidence

GROUP OCR boxes across documents by approximate position

FOR each position cluster:
  compute text repetition rate
  compute normalized text similarity
  compute visual/style consistency

  IF repetition_rate >= 0.85:
    label as CONSTANT
  ELSE IF value belongs to controlled lookup list:
    label as CONTROLLED_VARIABLE
  ELSE:
    label as OPEN_VARIABLE candidate

```

8.2 Output of Baseline

The baseline outputs:

- constant template regions,
- candidate variable fields,
- candidate controlled fields,
- uncertain regions,
- weak labels for training,
- logs for human review.

8.3 Why This Works Early

The baseline gives:

- interpretable behavior,
- easy debugging,
- usable Week-1/Week-2 extraction,
- weak labels for a later token classifier,
- template drift monitoring.

9. Marksheet Table Extraction Path

Marksheets are table-heavy, so they need a dedicated table path.

9.1 Recommended Default Model

Use PaddleOCR PP-Structure / table structure recognition as the first default because it fits naturally with the OCR stack and avoids introducing a separate table-recognition framework mid-sprint. Its table recognition documentation describes row, column, and cell-position recognition for converting table images into structured formats such as HTML.²

Optional later alternatives:

- Microsoft Table Transformer / TATR.
- LayoutParser + custom row/column grouping.
- Fine-tuned table detector if enough marksheets exist.

9.2 Marksheet Algorithm

INPUT marksheet

```
-> preprocess image
-> OCR full page
-> detect table regions
-> run table-structure recognition
-> recover rows, columns, and cells
-> map cells to headers
-> classify cells as CONTROLLED_VARIABLE or OPEN_VARIABLE
-> validate subject codes and grade patterns
-> export structured JSON
```

9.3 Table JSON Output

```
{
  "schema_version": "1.0",
  "document_type": "marksheet",
  "tables": [
    {
      "table_type": "subject_marks",
      "rows": [
        {
          "subject_code": "CS501PC",
          "subject_name": "Database Management Systems",
          "internal_marks": "24",
          "external_marks": "68",
          "total_marks": "92",
          "grade": "0",
          "credits": "3"
        }
      ]
    }
  ]
}
```

²PaddleOCR table-structure-recognition documentation: https://paddlepaddle.github.io/PaddleOCR/main/en/version3.x/module_usage/table_str

```

    }
  ]
}
]
}

```

10. Model Strategy

Do not train many models in parallel at the beginning.

10.1 MVP Model Path

Phase order:

1. Template-differencing baseline.
2. OCR + rules + lookup validation.
3. One token/layout classifier if enough annotation exists.
4. PP-Structure table path for marksheets.
5. Optional visual detector for seal/logo/signature only if needed.

10.2 Token/Layout Classifier

Possible models:

- LayoutLMv3.
- LayoutXLM if multilingual layout is needed.
- Donut-style OCR-free model only after OCR baseline is understood.

Labels:

- B-CONSTANT / I-CONSTANT.
- B-CONTROLLED_VARIABLE / I-CONTROLLED_VARIABLE.
- B-OPEN_VARIABLE / I-OPEN_VARIABLE.
- B-TABLE_HEADER / I-TABLE_HEADER.
- B-TABLE_CELL / I-TABLE_CELL.
- B-SEAL / I-SEAL.
- B-SIGNATURE / I-SIGNATURE.
- B-LOGO / I-LOGO.

11. Confidence Calibration

Do not multiply scores like:

```
field_confidence = token_confidence * ocr_confidence * layout_confidence
```

This is miscalibrated and produces misleading confidence values.

Instead, log component scores:

- OCR confidence.
- Token classifier confidence.
- Layout/template confidence.
- Repetition rate.
- Controlled-vocabulary match score.
- Table cell confidence.
- Quality score.

Then fit a calibration model:

```
features -> logistic regression / isotonic regression -> calibrated correctness probability
```

Calibration target:

Was this extracted field actually correct after human review?

12. Evaluation Plan and Numeric MVP Targets

The headline metric should be template-holdout performance, not random split performance.

12.1 Evaluation Splits

Split	Purpose	Priority
Template holdout	Tests generalization to unseen layout/template version	Headline
Document-type holdout	Tests generalization across certificate/marksheet style	Secondary
Institution holdout	Future test when more institutions exist	Future
Random split	Gives ceiling estimate only	Secondary

12.2 MVP Targets

Metric	MVP Target	Stretch Target
Template-holdout field F1	≥ 0.80	≥ 0.88
OPEN_VARIABLE field F1	≥ 0.82	≥ 0.90
CONTROLLED_VARIABLE accuracy	≥ 0.90	≥ 0.96
CONSTANT region precision	≥ 0.92	≥ 0.97
Marksheet table cell extraction F1	≥ 0.75	≥ 0.85
OCR character error rate on clean English text	$\leq 8\%$	$\leq 4\%$
Human review routing precision	≥ 0.75	≥ 0.85
JSON schema validity	100%	100%

13. Output Schema

Every output must include a schema version.

```
{
  "schema_version": "1.0",
  "project": "P1_FIELD_EXTRACTION",
  "document_id": "DOC_00042",
  "institution": "JNTUH",
  "document_type": "degree_certificate",
  "template_id": "JNTUH_CERT_V2",
  "fields": {
    "student_name": {
      "value": "A. Ravi Kumar",
      "label_class": "OPEN_VARIABLE",
      "bbox": [241, 318, 468, 351],
      "confidence": 0.94,
      "requires_review": false
    },
    "degree": {
      "value": "Bachelor of Technology",
      "label_class": "CONTROLLED_VARIABLE",
      "validated_against": "degree_lookup_v1",
      "confidence": 0.98,
    }
  }
}
```

```

    "requires_review": false
  },
  "quality": {
    "blur_score": 0.11,
    "skew_deg": 1.2,
    "low_contrast": false
  }
}

```

14. Template Drift Monitoring

Template drift means the institution quietly changes the layout, logo, seal placement, font, or wording.

Monitor:

- new repeated box positions,
- low template match rate,
- new controlled values,
- sudden drop in extraction confidence,
- rising human review rate.

If drift is detected:

```

mark template as POSSIBLE_NEW_VERSION
route sample documents to admin review
create new template version only after approval

```

15. Privacy and Compliance

This system processes student records and must be privacy-first.

The privacy design should explicitly align with India's Digital Personal Data Protection Act, 2023, which provides a framework for processing digital personal data while recognizing individuals' right to protect their personal data.³

Core requirements:

- Consent or valid processing basis where required.
- Purpose limitation.
- Data minimization.
- Role-based access control.
- Encryption at rest and in transit.
- Audit logging.
- Secure deletion.
- Breach-response workflow.
- No unnecessary long-term storage of raw student documents.

16. Six-Week Roadmap for P1

Week 1 - Data and Shared Core

Tasks:

- Collect/scan 50-100 real documents.
- Annotate at least 20 documents.
- Prepare 5-10 marksheet table annotations.
- Build upload, preprocessing, quality scoring, and OCR pipeline.

³Ministry of Electronics and Information Technology, Digital Personal Data Protection Act, 2023: <https://www.meiti.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf>

- Store OCR boxes in a common schema.

Done when:

- 50 documents are ingested.
- 20 documents have field annotations.
- OCR boxes and quality scores are generated for all documents.

Week 2 - Template-Differencing Baseline

Tasks:

- Normalize OCR boxes by page size.
- Cluster boxes by position.
- Compute repetition rates.
- Add CONSTANT / CONTROLLED_VARIABLE / OPEN_VARIABLE rules.
- Build lookup lists for degree, branch, semester, regulation, subject code.

Done when:

- System extracts 10-15 key fields from degree certificates.
- Output includes interpretable labels and confidence components.

Week 3 - Marksheet Table Path

Tasks:

- Integrate PaddleOCR PP-Structure table recognition.
- Extract rows, columns, and cells.
- Map cells to headers.
- Export marksheet tables to JSON.

Done when:

- At least one marksheet template produces structured subject-wise JSON.
- Table cell extraction F1 reaches at least 0.75 on the annotated subset.

Week 4 - Token/Layout Model Optional MVP Extension

Tasks:

- Train or fine-tune one token/layout classifier if annotation volume is sufficient.
- Compare against baseline.
- Keep baseline if model does not improve template-holdout score.

Done when:

- Template-holdout field F1 is measured.
- System chooses baseline or model based on validation performance, not assumption.

Week 5 - Calibration and Review Queue

Tasks:

- Collect human review correctness labels.
- Train logistic/isotonic confidence calibration.
- Build review queue for low-confidence fields.
- Add error taxonomy and structured logs.

Done when:

- Extracted fields have calibrated confidence.
- Review items include reason codes.

Week 6 - Evaluation and Demo

Tasks:

- Run template-holdout evaluation.
- Run OCR benchmark.
- Run marksheet table evaluation.
- Build final dashboard/API demo.
- Prepare final report.

Done when:

- MVP targets are reported.
- Demo shows certificate extraction and marksheet table extraction.
- JSON outputs pass schema validation.

17. Final Recommended Build Order

1. Data collection and annotation
2. OCR/preprocessing core
3. Template-differencing baseline
4. Controlled vocabulary validation
5. Marksheet table extraction
6. Confidence calibration
7. Human review queue
8. Optional token/layout classifier
9. Final evaluation and demo

18. Final P1 Definition

P1 is a privacy-aware academic document extraction system that classifies document content into `CONSTANT`, `CONTROLLED_VARIABLE`, and `OPEN_VARIABLE` regions, extracts structured fields from degree certificates and marksheets, reconstructs marksheet tables using a dedicated table-structure module, calibrates confidence using validation data, and reports template-holdout performance as the main success metric.